

Finding Principal Directions

Mathematics of Machine Learning

Math 3180 — University of Connecticut

The Data Matrix

- ▶ X is an $N \times k$ matrix:
 - ▶ rows $\longleftrightarrow$ samples
 - ▶ columns $\longleftrightarrow$ features
- ▶ **Assumption:** the data is *centered* — each column sums to zero.
- ▶ Each row of X is a point in $\mathbb{R}^k$.
The N rows form a *cloud of points* in $\mathbb{R}^k$.

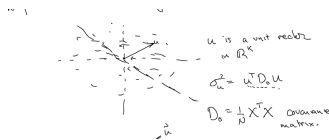
Variance in a Direction

- ▶ Let $u \in \mathbb{R}^k$ be a **unit vector**, $\|u\|^2 = 1$.
- ▶ The *variance of the data in direction u* is

$$\sigma_u^2 = u^T D_0 u,$$

where $D_0 = \frac{1}{N} X^T X$ is the **covariance matrix**.

- ▶ $\sigma_u^2 = \frac{1}{N} \sum_{i=1}^N (x_i \cdot u)^2$ is the mean squared projection onto u .



The Optimization Problem

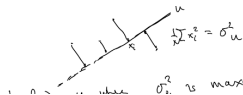
Goal

Find the unit vector $u \in \mathbb{R}^k$ that **maximizes** $\sigma_u^2 = u^T D_0 u$.

- This is a *constrained* optimization problem:

$$\text{maximize } u^T D_0 u \quad \text{s.t.} \quad \|u\|^2 = 1.$$

- Tool: **Lagrange multipliers**.



Projections $x_i \cdot u$ of data points onto u ; their mean square is σ_u^2 .

Lagrange Multipliers: General Setup

- ▶ **Objective:** $F(x_1, \dots, x_k)$ — the function to maximize.
- ▶ **Constraint:** $g(x_1, \dots, x_k) = 0$.

Form the **Lagrangian**:

$$S(x_1, \dots, x_k, \lambda) = F - \lambda g.$$

Critical points satisfy:

$$\frac{\partial S}{\partial x_i} = 0 \quad (i = 1, \dots, k) \quad \text{and} \quad \frac{\partial S}{\partial \lambda} = 0 \iff g = 0.$$

Setting Up the Lagrangian

Write $D_0 = (d_{ij})$. Then

$$\sigma_u^2 = u^T D_0 u = \sum_{r=1}^k \sum_{s=1}^k u_r d_{rs} u_s.$$

The constraint is $g(u) = u_1^2 + \cdots + u_k^2 - 1 = 0$.

The Lagrangian is:

$$S(u, \lambda) = \sum_{r=1}^k \sum_{s=1}^k u_r d_{rs} u_s - \lambda(u_1^2 + \cdots + u_k^2 - 1).$$

Computing the Partial Derivatives

Differentiating S with respect to u_i and using the symmetry $d_{rs} = d_{sr}$ of D_0 :

$$\frac{\partial S}{\partial u_i} = 2 \sum_{s=1}^k d_{is} u_s - 2\lambda u_i.$$

In vector form:

$$\nabla_u S = 2 D_0 u - 2\lambda u = 2(D_0 - \lambda I) u.$$

The condition $\frac{\partial S}{\partial \lambda} = 0$ gives $\|u\|^2 = 1$.

The Critical Point Equations

Setting $\nabla_u S = 0$ and $\partial S / \partial \lambda = 0$ gives the system:

$$\begin{cases} (D_0 - \lambda I) u = 0, \\ \|u\|^2 = 1. \end{cases}$$

The first equation is equivalent to

$$\boxed{D_0 u = \lambda u.}$$

u is an eigenvector of D_0 with eigenvalue λ , normalized so that $\|u\| = 1$.

The Value of σ_u^2 at a Critical Point

If $D_0 u = \lambda u$ and $\|u\| = 1$, then

$$\sigma_u^2 = u^T D_0 u = u^T (\lambda u) = \lambda \|u\|^2 = \lambda.$$

The variance in direction u *equals* the corresponding eigenvalue.

Main Theorem

Theorem

The critical points of $\sigma_u^2 = u^T D_0 u$ subject to $\|u\|^2 = 1$ are exactly the unit eigenvectors of D_0 . At such a critical point, $\sigma_u^2 = \lambda$ (the corresponding eigenvalue).

- ▶ The **maximum variance** equals the **largest eigenvalue** of D_0 ; it is achieved along the corresponding eigenvector.
- ▶ The **minimum variance** equals the **smallest eigenvalue** of D_0 .

Fact

$D_0 = \frac{1}{N} X^T X$ is symmetric, so all its eigenvalues are real (and in fact non-negative).